

RUDRA RAJ NARAYAN MONAS

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Education

Indian Institute of Information Technology Vadodara <i>B.Tech in Computer Science and Engineering</i> CPI: 8.98	Aug 2023 – May 2027 <i>Gandhinagar, Gujarat, India</i>
Kendriya Vidyalaya Ujjain <i>Class XII (CBSE) – 93.80%</i> — <i>Class X (CBSE) – 94.60%</i>	Apr 2019 – Mar 2022 <i>Ujjain, Madhya Pradesh</i>

Experience

IIT Kanpur <i>SURGE Research Intern — Perception and Intelligence Lab</i>	May 2026 – Present <i>Kanpur, Uttar Pradesh</i>
- Designed a novel zero-shot denoising method for microscopic images targeting Gaussian and Poisson noise. - Implemented blind-spot CNN-based denoising models in PyTorch/TensorFlow with CUDA-accelerated training. - Benchmarked model performance against existing state-of-the-art methods on FMD and W2S microscopy datasets.	
IIIT Vadodara <i>Teaching Assistant — Data Structures Lab</i>	Jan 2026 – Apr 2026 <i>IIIT Vadodara</i>
- Assisted laboratory sessions for 180+ students implementing data structures and algorithms in C/C++. - Evaluated assignments, viva examinations, and supported students with debugging and complexity analysis.	
InventX Accelerator, IIT Gandhinagar <i>Product Innovation Intern</i>	May 2025 – Jul 2025 <i>Internship, Gandhinagar, Gujarat</i>
- Designed a vibration-damping handle grip using signal processing and Fourier Transform-based vibration analysis. - Prototyped and tested designs using Arduino, Fusion 360/SolidWorks, 3D printing, and laser cutting. - Reduced measured hand-transmitted vibration levels from 3.7 m/s² to 2.1 m/s² through iterative design and testing.	

Projects

Zero-Touch NWDAF <i>Python, PyTorch, Ella Core, UERANSIM</i> GitHub
- Developed a framework for real-time 5G signaling anomaly detection and autonomous mitigation. - Implemented an LSTM Autoencoder, ABMM mobility model, and a complete Ella Core+UERANSIM testbed. - Achieved 0.989 Precision, 1.000 Recall, 0.994 F1-score, and 2 s detection latency.
FEDSEG-APP <i>Python, PyTorch, FastAPI, Streamlit, Render, Federated Learning</i> GitHub
- Developed a federated chest X-ray segmentation platform using FastAPI, Streamlit, and ResNet34-U-Net. - Improved Dice score from 0.7429 to 0.7927 on QaTa-COV19 using prototype-guided learning & uncertainty filtering.
Distributed Access Control in Pervasive Edge Computing <i>C/C++, Cryptography, Edge Computing, Bilinear Pairings</i>
- Developed a decentralized access-control framework for secure edge-computing data sharing. - Implemented Key-Aggregate Cryptography, proxy re-encryption, and challenge-response authentication. - Achieved 32 ms latency with 2.72 KB communication overhead per authentication round.

Achievements

- Won Second Best Invention at InventX'25, securing INR 1,00,000 and filing a provisional patent. - Achieved peak ratings of 1533 on CodeChef and 1506 on LeetCode. - Qualified GATE 2026 (Computer Science) with AIR 6271 and earned NPTEL Elite+Silver (Top 5%) in Machine Learning.

Technical Skills

Languages: C, C++, Python, MATLAB, SQL, Bash Core CS: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP ML & AI: PyTorch, TensorFlow, CUDA, Computer Vision, Image Processing, Federated Learning, Deep Learning Frameworks & Tech.: FastAPI, Streamlit, REST APIs, OpenSSL, UERANSIM, Ella Core Tools: Git, GitHub, Linux, Docker

Leadership & Extracurricular Activities

Student Placement Committee (SPC) <i>Student Point of Contact (POC)</i>	Jan 2026 – Present <i>IIIT Vadodara</i>
Served as student liaison for placement communication, recruitment drives, and career-readiness workshops.	
Academic Committee <i>Core Member</i>	July 2024 – April 2025 <i>IIIT Vadodara</i>
Organized academic and cultural activities, helping conduct 10+ events for 300+ participants.	